Erfan Pirjahandideh

Computer vision and applied machine learning

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github.com/erfanpj | LinkedIn | Project portfolio

Profile

Computer vision and AI research assistant at Fraunhofer IGD and AI master's student at TU Darmstadt. I bring eight years in telecommunications IT to research on how models behave outside their training data. I like turning a promising idea into something another person can inspect, question and use.

How I work

I notice where a workflow breaks down, find the question behind it and choose one improvement to test. I build a small version, compare it with a simple baseline, then share both the result and its limits. In my EEG project, that meant showing how scores varied by participant alongside the overall average.

Technical skills

Computer vision and generative models. Python, PyTorch, CUDA, OpenCV, Hugging Face Diffusers and Datasets; Stable Diffusion, pretrained LoRA, face alignment and biometric embeddings.

Evaluation and reproducibility. Participant-held-out testing, baseline comparisons, EER and FMR/FNMR, experiment logging, identity-wise sampling and SHA-256 dataset checks. Git, Linux, Docker and Slurm.

Software and operations. Django, APIs, PostgreSQL, PHP, Laravel and JavaScript. Service monitoring, Jira integration, workflow automation, software testing and technical documentation.

Selected independent projects

Brain Signals Lab Demo

Evaluated imagined left- versus right-hand movement on public EEG data from 50 people. Across five participant-held-out folds, CSP + LDA averaged 61.8% accuracy against a 50.4% most-frequent-class baseline. The participant median was 57.8%; the viewer exposes the variation.

OpsDesk Demo

Built a fictional support desk in which revising a draft cancels its approval. On 60 held-out fictional tickets, rules made 37 fully correct proposals; a local language model made 25. The browser demo shows the approval flow with simulated actions.

Reachboard Demo

Built a Django campaign workflow with fixed quotes, advertiser-owned drafts and repeat submissions that leave the audit history unchanged. A separate browser demo lets visitors try the workflow using fictional placements and prices.

I used AI coding assistance on these personal projects.

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Professional experience

Computer Vision and AI Research Assistant

Fraunhofer IGD, Darmstadt, Germany | February 2026–present

Build reproducible identity-synthesis and evaluation pipelines with Python, PyTorch, CUDA and Stable Diffusion. Align faces, extract 512-dimensional embeddings and compare biometric error rates across controlled attribute experiments.

Run experiments on A100 GPUs with Slurm. Apply quality scoring and identity-wise selection to training data, verify dataset integrity, and document evaluation procedures. Developed forms and review checklists for dental data acquisition.

Telecommunications IT

NAK and Behsa, MCCI projects; AHS, MTN project | 2016–2024

NAK. Led implementation of monitoring for more than 30 services and developed a Chrome extension for ticket handling.

Behsa. Led a seven-person service desk handling more than 2,000 tickets a month. Integrated the MCI ticketing system with Jira and implemented ITIL-aligned workflows.

AHS. Automated management reporting and developed ticketing extensions and service desk dashboards.

Freelance software development and web design

Independent | 2005–present

Develop responsive websites and web applications using PHP, WordPress, Joomla and JavaScript. Use AI coding tools for implementation, debugging and documentation.

Education

M.Sc. Artificial Intelligence and Machine Learning

TU Darmstadt | 2024–present

Degree in progress; expected completion in the first half of 2027.

Software engineering and earlier AI studies

Islamic Azad University

B.Sc. and Associate Degree in Software Engineering. Earlier master's study in AI: 20 credits completed; no thesis or degree awarded.

Publication

Co-author, "Personality Recognition from Text Based on Deep Learning with Recurrent Neural Network." 5th International Conference on Fundamental Research in Computer Engineering and Information Technology, 2017.

Award and training

MTN CEO Special Award for Innovation, April 2018. Training: ISTQB software testing and DevOps Foundation (2021); Python Foundation and ITIL Foundation (2017).

Languages

Farsi: native | English: advanced | German: A1